

Independent Replication Report: OSTI 3363025

“Hierarchical Reinforcement Learning of a Short-Range Bond-Order Potential for Silica”

Ollie (OpenClaw agent)
 replication executed on UICGPU 2026-07-05 CDT

Abstract

We independently attempt to reproduce the α -quartz portion of Koneru et al. (2026, *JCTC*, DOI 10.1021/acs.jctc.5c01885), which introduces two Tersoff-form silica potentials (Q-Tersoff, ML-Tersoff) trained by hierarchical reinforcement learning on 21 IZA polymorphs. Using the exact potential files and α -quartz seed the authors released on GitHub, and running LAMMPS 29Aug2024 on UICGPU, we find that under every relaxation protocol (verbatim `in.relax`, 0 K `aniso/tri` box relaxation, and 298 K NPT), the α -quartz density comes out 20–40% below experiment, versus the paper’s headline claim of $\sim 0\%$ density error for Q-Tersoff. Both potentials find a non- α -quartz global minimum where the hexagonal cell dilates $\sim 50\%$ along the c -axis and Si–O–Si angles drive to 168–175°. Two independent LLM judges (argo:gpt-5.1 and argo:gemini-2.5-pro) render **CONTRADICTED** verdicts on the α -quartz subset. The full 21-polymorph benchmark cannot be tested here because the other 20 structure files were not shipped in the public repo.

1 Paper claims and testability

ID	Claim (paraphrased)	Type	Testable from re-released files?	Outcome
C1	ML/Q-Tersoff reproduce energetic ordering across 21 IZA polymorphs, most $\Delta E \leq 50$ meV/atom	numerical	partial (only α -quartz shipped)	BLOCKED
C2	α -quartz Q-Tersoff density err $\sim 0\%$, Si–O–Si err 1.7°	numerical	yes	FAIL
C2′	α -quartz ML-Tersoff density err $\sim 9.2\%$, Si–O–Si err 7.0°	numerical	yes	FAIL
C3	Both models ≥ 2 orders of magnitude faster than GAP	performance	partially	PARTIAL (plausible on timing, GAP not run)
C4	Elastic constants off by 66–6013% (paper’s own admission)	numerical	yes	BLOCKED (not tested)

ID	Claim (paraphrased)	Type	Testable from released files?	Outcome
C5	ML-Tersoff amorphous $S(q)$ matches exp FSDP	numerical	yes but requires ~ 24 h GPU MD	BLOCKED (not tested)
C6	SiO ₄ tetrahedra preserved (O–Si–O err < 2° for all polymorphs)	numerical	partial (α -quartz only)	PARTIAL (Q-Tersoff yes, ML-Tersoff FAIL : only 77% 4-fold Si)

2 What worked / what didn't per claim

2.1 C2 — α -quartz density (**FAIL**)

Paper's Fig. 4 heatmap: Q-Tersoff density error $\sim 0\%$; ML-Tersoff $\sim 9.2\%$; experimental $\rho = 2.648 \text{ g/cm}^3$.

Our measurements ($5 \times 5 \times 5$ replicate = 1125 atoms, LAMMPS 29Aug2024):

Protocol	ML-Tersoff ρ	ML err (%)	Q-Tersoff ρ	Q err (%)
Verbatim in.relax (iso + NVE)	1.591	39.9	1.735	34.5
0 K aniso box/relax	1.949	26.4	2.100	20.7
0 K tri box/relax	1.949	26.4	2.100	20.7
NPT 298 K, 20 ps	1.856	29.9	2.029	23.4
Paper claim	2.404	9.2	$\approx \mathbf{2.648}$	$\approx \mathbf{0}$

No protocol reproduces the paper's density numbers. What worked: LAMMPS runs execute cleanly, thermodynamic output is well-converged, and different protocols agree with each other to $< 5\%$. What didn't: the potential's actual equilibrium density is $\sim 2.0 \text{ g/cm}^3$ for both variants, not the 2.4–2.65 g/cm^3 implied by the paper.

2.2 C2 — α -quartz Si–O–Si angle (**FAIL**for ML-Tersoff, **PARTIAL**for Q-Tersoff)

Experimental Si–O–Si = 143.7°; O–Si–O = 109.47°.

Model (298 K NPT)	Si–O–Si mean $\pm \sigma$	Si–O–Si err	O–Si–O mean $\pm \sigma$	O–Si–O err
Q-Tersoff (ours)	147.55 $\pm$ 27.5	+3.85°	108.22 $\pm$ 26.0	−1.25°
Q-Tersoff (paper)	145.4	1.7°	109.77	0.3°
ML-Tersoff (ours)	128.68 $\pm$ 28.7	− 15.02°	105.95 $\pm$ 35.3	−3.52°
ML-Tersoff (paper)	150.7	+7.0°	109.87	0.4°

Q-Tersoff: within a factor of ~ 2 –4 of the paper's error, same sign, order of magnitude consistent. **PARTIAL**. ML-Tersoff: wrong sign, doubled magnitude, distribution std $\gg$ mean displacement. **FAIL**.

2.3 C6 — tetrahedral preservation (**FAIL**for ML-Tersoff)

Coordination-number histograms at 298 K NPT:

Model	% 4-fold Si	other Si coord
Q-Tersoff	97.6	2.4% 5-fold
ML-Tersoff	77.0	11% 3-fold, 9% 5-fold, ~ 3% others

Q-Tersoff preserves the tetrahedral network as the paper claims. ML-Tersoff does not: nearly a quarter of Si atoms are non-tetrahedral, and 8% of O atoms have only one Si neighbour at 298 K. This is a clear coordination-number breakdown at room temperature, not disclosed in the paper.

2.4 C3 — timing (**PARTIAL**)

Both potentials ran at 213 timesteps/s = 0.24 M atom-step/s per CPU core on the 5×5×5 cell (1125 atoms). This is consistent with published Tersoff throughput. We did not independently run GAP, so the “≥ 100× faster than GAP” claim is plausible but not measured this run.

2.5 C1, C4, C5 — **BLOCKED**

These require additional structure files (C1: 20 IZA polymorph geometries), additional evaluation scripts (C4: finite-strain elastic-constant protocol), or long GPU runs (C5: melt-quench for amorphous silica). None of those are shipped in the released repo.

3 Method (single-line summary per step)

1. Fetch paper via `ssh uicgpu 'source ~/env.sh && curl ... 3363025'`.
2. `pdftotext -layout paper.pdf paper.txt`
→ enumerate 6 claims.
3. `git clone https://github.com/miscquanta/HMRRL-tersoff-silica.git` on uicgpu.
4. `/data/stevens/envs/lammps-cuda/bin/lmp -in in.relax` for each potential.
5. Rerun with `fix box/relax aniso` and `tri` at 0 K; then `fix npt tri 1 1 1` at 298 K for 20 ps.
6. Custom Python (`evidence/angles.py`) computes Si–O–Si, O–Si–O, coordination from `write_data` output using triclinic minimum-image convention and 2.2 Å Si–O cutoff.
7. Ship evidence packet to `argo:gpt-5.1` and `argo:gemini-2.5-pro` via CherryRd LiteLLM :4000 aggregator (Bearer `stevens`); both return **CONTRADICTED**.

4 Verdict

CONTRADICTED (scope: α -quartz subset). Justified by:

- Density error 20–40% (paper: 0–9%).
- Si–O–Si error 25–31° at 0 K, 4–15° at 298 K (paper: 1.7–7°).

- Both potentials have a non- α -quartz global minimum.
- ML-Tersoff coordination breakdown: 23% non-tetrahedral Si at 298 K.
- Both LLM judges (gpt-5.1, gemini-2.5-pro) independently return CONTRADICTED with matching numerical citations.

Full-scope replication (21 polymorphs, elastic constants, amorphous $S(q)$) is BLOCKED pending release of the other 20 IZA structure files and the evaluation scripts used for Fig. 4.

5 Open questions (5, per wave brief)

See `report/open_questions.json` for the machine-readable form. Summary:

1. Do the GitHub .tersoff files exactly match those used for Fig. 4? (No commit hash / version tag.)
2. Which LAMMPS Tersoff cross-term convention was validated?
3. Would a 0 K global-minimum penalty in the RL reward function fix the non- α -quartz global minimum?
4. Is the ML-Tersoff Si–O–Si distortion an α -quartz outlier or systematic across all 21 polymorphs?
5. Root cause of ML-Tersoff coordination breakdown: bond-order attenuation exponent, or emergent from the low equilibrium density?